

Style360: AI-Powered Virtual Try-On Platform

Project Proposal

1. Overview

Style360 is a next-generation Virtual Try-On web platform that creates a personalized, 360° 3D avatar from basic 2D photos to let users layer clothes and get real-time AI styling advice. Remarkably, the entire system is built to run at zero cost by utilizing open-source models, local XAMPP hosting, and free-tier APIs like Hugging Face and Google Gemini.

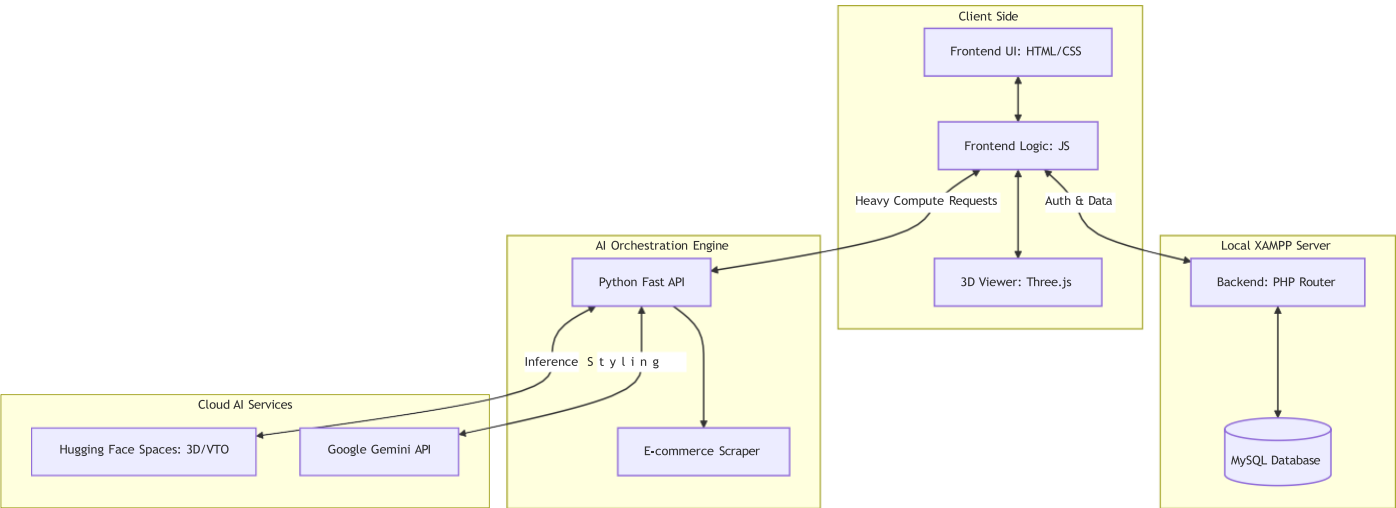
2. Challenges Intended to Solve Using AI Integration

E-commerce fashion currently suffers from high return rates and poor user engagement due to the inability to physically try on clothes. Style360 leverages AI to solve these specific challenges:

- **Fit Heatmap:** Extracts 2D biometrics into a 3D heatmap, showing where clothes fit or pinch. Lack
- **of Visualization:** Maps garments onto a personalized 3D avatar to view from any angle.
- **Absence of Personal Styling:** Provides 24/7 personalized outfit and color recommendations matching the user's skin tone and body shape.

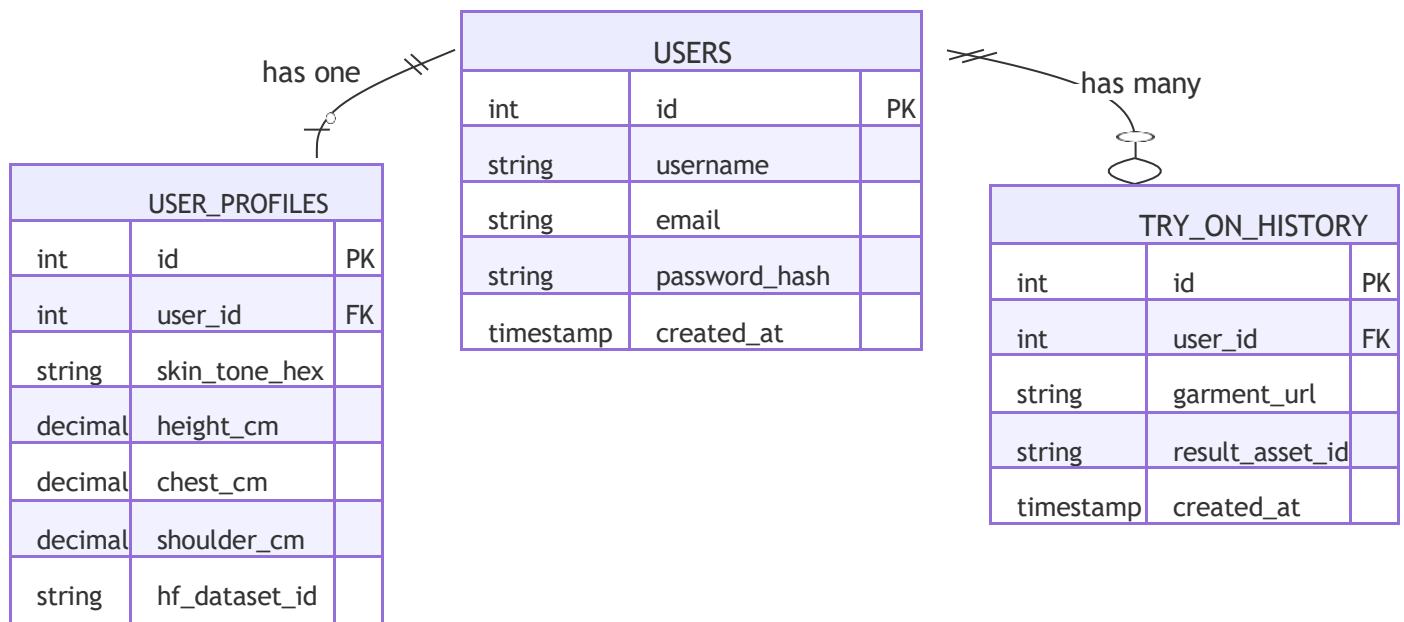
3. System Architecture Diagram

The platform utilizes hybrid architecture, separating the lightweight frontend and PHP backend from the heavy AI computation engine.



4. System ER Diagram

The database schema is designed to manage user identities, store their AI-extracted biometrics, and keep a history of their virtual try-on sessions.



5.Functional Requirements

1. **Account Management**-Users can securely register, log in, and save their profiles.
2. **Avatar Creation**- Processes an uploaded image to extract biometrics and generate a custom 3D mesh.
3. **Interactive Viewer**- Renders the 3D avatar with 360-degree rotation and focused zoom controls.
4. **Clothing Overlay**-Applies garments to the avatar via an uploaded image or product URL.
5. **Chat Assistant**- Features a Gemini-powered chat interface for real-time styling advice.

6.Non-Functional Requirements

1. **Cost Efficiency**: \$0 operational cost using open-source tech and free APIs.
2. **Usability**: Modern, premium UI aesthetics with intuitive controls.
3. **Availability**: Handles free-tier API delays gracefully with loading indicators.
4. **Portability**: Easily deployable on a standard local XAMPP environment.

7.Technology Details

- **Frontend**: Vanilla HTML5, CSS3, and JavaScript.
- **3D Rendering**: Three.js with Orbit Controls for browser-based 3D avatar and heatmap display.
- **Backend**: PHP via a local XAMPP server for auth, sessions, and CRUD actions.
- **Database**: MySQL for relational data storage.
- **AI Orchestration**: Python (Fast API) microservice for heavy logic, BeautifulSoup scraping, and API routing.
- **AI Models**: Hugging Face Spaces for 3D reconstruction and try-on without needing local GPUs.
- **Styling Assistant**: Google Gemini API (Free Tier) for natural language styling advice.

8.Conclusion

Style360 bridges the gap between flat e-commerce browsing and the physical fitting room. By combining lightweight web tech (PHP/JS) with powerful cloud-based AI (Python/Hugging Face/Gemini), it delivers a premium, highly personalized user experience. Crucially, the strategic use of open-source tools ensures the platform operates at zero infrastructure costs, making it both technologically advanced and highly accessible.

9. GitHub Link- <https://github.com/anjana20033941/Style360>